

JavaScript Fundamentals: Input, Output, Arrays, and Logic Reference Guide

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Execution Environment

This guide is designed for the **Browser Console**. To run these scripts, open Google Chrome or Firefox, right-click anywhere, select **Inspect**, navigate to the **Console** tab, paste your code, and hit Enter.

1 Essential JavaScript Coding Elements

1.1 Inputs and Outputs

When practicing JS without HTML, we use `prompt()` to take inputs from the user and `console.log()` to display the output. **Note:** `prompt()` always returns a String. If you need a number, you must convert it using `Number()` or `parseInt()`.

```
1 // Outputting a message
2 console.log("Welcome to JavaScript!");
3
4 // Taking string input
5 let userName = prompt("What is your name?");
6 console.log("Hello, " + userName);
7
8 // Taking numeric input
9 let ageInput = prompt("Enter your age:");
10 let age = Number(ageInput); // Conversion
11 console.log("You will be " + (age + 1) + " next year.");
```

1.2 Variables and Data Types

Variables store data. Use `const` for values that never change, and `let` for values that will change.

```
1 const pi = 3.14159; // Number (Constant)
2 let score = 0; // Number (Variable)
3 let isGameOver = false; // Boolean
4 let playerName = "John"; // String
```

1.3 Operators (Arithmetic & Logical)

Operators are used to perform calculations or evaluate conditions.

```
1 // Arithmetic
2 let sum = 10 + 5;
3 let remainder = 10 % 3; // Modulus (Returns 1)
4
5 // Comparison and Logical
6 let isAdult = (age >= 18);
7 let hasTicket = true;
8 let canEnter = (isAdult && hasTicket); // Logical AND
```

1.4 Control Flow (If/Else)

Used to make decisions in your code based on conditions.

```
1 let marks = Number(prompt("Enter your marks:"));
2
3 if (marks >= 80) {
4     console.log("You got an A+!");
5 } else if (marks >= 40) {
6     console.log("You passed.");
7 } else {
8     console.log("You failed.");
9 }
```

1.5 Loops

Loops repeat a block of code until a certain condition is met.

```
1 // For Loop (ideal when you know how many iterations you need)
2 for (let i = 1; i <= 3; i++) {
3     console.log("Attempt number: " + i);
4 }
5
6 // While Loop (ideal for continuing until a state changes)
7 let count = 3;
8 while (count > 0) {
9     console.log("Countdown: " + count);
10    count--; // Decrement by 1
11 }
```

1.6 Functions

Functions encapsulate code so it can be reused multiple times.

```
1 // Defining a standard function
2 function calculateArea(width, height) {
3     return width * height;
4 }
```

```

5
6 // Using the function with prompt and console
7 let w = Number(prompt("Enter width:"));
8 let h = Number(prompt("Enter height:"));
9 let area = calculateArea(w, h);
10 console.log("The area is: " + area);

```

1.7 Arrays: Basics and Modification

Arrays are special variables that hold multiple values. You can add, remove, and replace elements using built-in array methods.

```

1 let fruits = ["Apple", "Banana"];
2
3 // Adding elements
4 fruits.push("Mango"); // Adds to the end -> ["Apple", "Banana", "Mango"]
5 fruits.unshift("Kiwi"); // Adds to the beginning -> ["Kiwi", "Apple", "Banana", "Mango"]
6
7 // Removing elements
8 fruits.pop(); // Removes from the end -> ["Kiwi", "Apple", "Banana"]
9 fruits.shift(); // Removes from the beginning -> ["Apple", "Banana"]
10
11 // Splice: array.splice(startIndex, deleteCount, newItem...)
12 fruits.splice(1, 1, "Orange"); // Replaces "Banana" with "Orange" -> ["Apple", "Orange"]

```

1.8 Advanced Array Methods (Map, Filter, Reduce)

These are powerful methods that loop through arrays and return new data without changing the original array.

```

1 let nums = [10, 20, 30];
2
3 // 1. map(): Transforms every element
4 let doubled = nums.map(n => n * 2);
5 console.log(doubled); // Output: [20, 40, 60]
6
7 // 2. filter(): Keeps elements that pass a true/false test
8 let over15 = nums.filter(n => n > 15);
9 console.log(over15); // Output: [20, 30]
10
11 // 3. reduce(): Boils the array down to a single value
12 let sum = nums.reduce((total, n) => total + n, 0);
13 console.log(sum); // Output: 60

```

Accessing Previous/Next Elements in Filter

When using methods like `filter()`, the callback provides three arguments: the current element, its index, and the original array. You can use the index to look backward or forward.

```
1 let temps = [20, 22, 21, 25];
2
3 // Keep days that are hotter than the previous day
4 let hotterDays = temps.filter((temp, index, array) => {
5   // Check boundaries: skip the first day because it has no
   // previous day
6   if (index === 0) return false;
7
8   let prevTemp = array[index - 1]; // Access the previous
   element
9   return temp > prevTemp;          // Keep if true
10 });
11
12 console.log(hotterDays); // Output:[22, 25]
```

2 Standard Practice Problems (With Solutions)

Problem 1: Temperature Converter

Task: Ask the user for a temperature in Celsius. Convert it to Fahrenheit and print the result.

```
1 let celsiusStr = prompt("Enter temperature in Celsius:");
2 let celsius = Number(celsiusStr);
3 let fahrenheit = (celsius * 9/5) + 32;
4 console.log(celsius + "C is equal to " + fahrenheit + "F.");
```

Problem 2: Even or Odd Checker

Task: Ask the user for a number. Print whether the number is Even or Odd.

```
1 let num = Number(prompt("Enter a number to check Even/Odd:"));
2 if (num % 2 === 0) {
3     console.log(num + " is an Even number.");
4 } else {
5     console.log(num + " is an Odd number.");
6 }
```

Problem 3: Simple ATM Logic (Functions & Flow)

Task: Initialize a balance of 5000. Ask the user if they want to 'withdraw' or 'deposit'. Ask for the amount. Warn them if they try to withdraw more than they have.

```
1 let balance = 5000;
2 let action = prompt("Type 'withdraw' or 'deposit':");
3 let amount = Number(prompt("Enter the amount:"));
4
5 function processTransaction(actionType, amt) {
6     if (actionType === "deposit") {
7         balance += amt;
8         console.log("Deposited: " + amt + ". New Balance: " +
9             balance);
10    } else if (actionType === "withdraw") {
11        if (amt > balance) {
12            console.log("Error: Insufficient funds!");
13        } else {
14            balance -= amt;
15            console.log("Withdrew: " + amt + ". New Balance: " +
16                balance);
17        }
18    }
19 }
20 processTransaction(action, amount);
```

Problem 4: Array Filter - Student Grades

Task: Given an array of grades, ask the user for a passing threshold. Use `filter()` to print only the passing grades.

```
1 let grades =[45, 82, 33, 90, 67, 50];
2 let passingScore = Number(prompt("Enter the passing score (e.g.,
   60):"));
3
4 let passingGrades = grades.filter(grade => grade >= passingScore)
   ;
5 console.log("Passing grades: " + passingGrades);
```

3 Additional Practice Exercises (Unsolved)

Challenge yourself by writing the JavaScript code for the following problems. Remember to use `prompt()` to gather inputs and `console.log()` to display your answers.

Problem 5: The Simple Calculator

Task: Ask the user for a **first number**, a **second number**, and a **math operator** (e.g., +, -, *, /). Use `if/else` to perform the correct operation and print the result.

Problem 6: Find the Largest Number

Task: Ask the user to input **three** different numbers using three separate `prompt()` commands. Write logic using logical operators (`&&`) to determine which is the largest.

Problem 7: Sum of N Numbers

Task: Ask the user to input a positive integer (N). Use a `for` loop to calculate the sum of all numbers from 1 up to N .

Problem 8: Leap Year Checker

Task: Ask the user for a year (e.g., 2024). Write a function that determines if it is a leap year (Divisible by 4. *Exception:* If divisible by 100, it is NOT a leap year, UNLESS it is also divisible by 400).

Problem 9: The FizzBuzz Challenge

Task: Ask the user for a number. Loop from 1 to that number. If divisible by 3, print "Fizz". If divisible by 5, print "Buzz". If divisible by BOTH, print "FizzBuzz". Otherwise, print the number.

Problem 10: Array Manipulation (Shopping List)

Task: Create an empty array called `shoppingList`. Use a `while` loop to continuously ask the user to type an item to buy (using `prompt()`).

- Add the item to the array using `push()`.
- If the user types "done", break the loop.
- Finally, print the entire array to the console.

Problem 11: Array Mapping (Currency Converter)

Task: You have an array of prices in Dollars: `[10, 25, 50, 100]`. Ask the user for the current exchange rate to BDT (e.g., 110). Use the `map()` method to create a new array with all prices converted to BDT. Print the new array.

Problem 12: Finding Valleys (Previous & Next Filter)

Task: You are given an array of altitudes: [500, 300, 400, 100, 200, 600]. A "valley" is a number that is strictly **less than** both the number before it AND the number after it. Use `filter()` and the index arguments to isolate and print only the valley numbers.

(Hint: Exclude the very first and very last indexes from your check, as they don't have both neighbors!)